BORGWARNER

Stall Burst Thermal Model

Revision 0.6

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1 Introduction

To improve functionality at stall and low speed, a thermal model has been implemented to allow for real time estimation of inverter switch junction temperature. This model allows for higher bursts of current from stall in a wide array of conditions while continuing to protect the inverter for long term use.

This methodology is implemented in the form of an inverter switch Junction Temperature model fed into a PI regulator targeting a max switch junction temperature. This regulator allows the inverter to have the full max burst current, as found by testing at Cascadia Motion, then the Junction Temperature Model and PI regulator derate that max burst current as necessary to maintain a max junction temperature.

Inverter	Target Max Switch Junction Temperature
CM200DX	210 °C for 652E and later
CM200DZ	149 °C
CM350DZ	149 °C

The Junction Thermal Model takes in: DC voltage in V_{DC}, switching frequency in Hz, motor speed in RPM, motor current in A_{pk}, and coolant temperature in °C, to predict the hottest inverter switch junction temperature in °C. Note: the model will change over time, but time is not an inherent input to the system other than the switching period.

The general architecture of control is presented below in Figure 1.

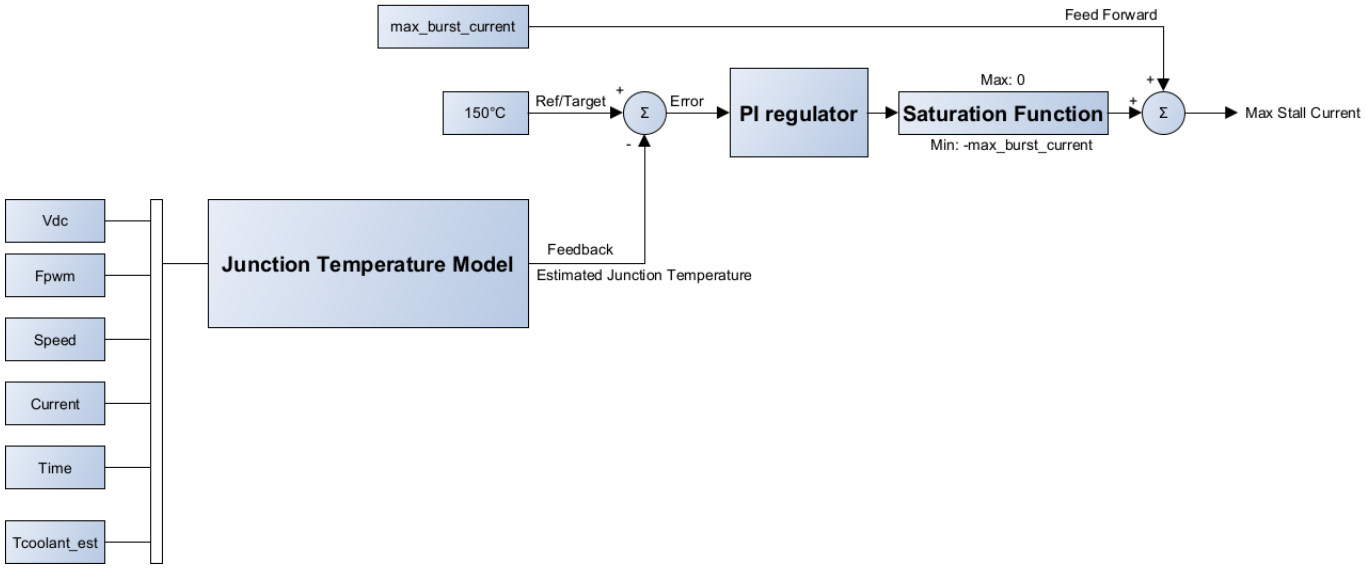


Figure 1: General architecture of Stall Burst Thermal Model.

Through this methodology, compared to a steady state model, more current is allowed at stall for a transient burst, assuming the inverter switch junction temperature is not already at its maximum switch junction temperature.

2 Implementation

2.1 Overview

The Stall Burst Thermal Model is implemented in the code base only for specific inverters. These inverters are listed in the table below. No user input is required to utilize the model and its resulting stall burst.

Inverter	Target Max Switch Junction Temperature
CM200DX	652E* and later
CM200DZ	6527 and later
CM350DZ	6527 and later

* Stall Burst Thermal Model was introduced in the CM200DX with firmware 6522. However, performance and details in this manual reflects 652E and later firmware for CM200DX.

2.2 Stall Burst Thermal Model Settings

There are no required user inputs to activate the Stall Burst Thermal Model outside of using the appropriate inverter.

However, it is highly recommended to limit the torque ramp rate of the system (via EEPROM Torque_Rate_Limit_EEPROM_(Nm)_x_10) to no more than 70 Nm/3ms or a setting of 700 for iM225 and iM375 integrated modules. This corresponds to a torque ramp rate of 23,000 Nm/s. Rates faster than this may cause stability issues.

There are three new signals relayed over CAN which report: the highest junction temperature by the model in °C, whether Burst Model Mode is in stall or not, and whether the Stall Burst Thermal Model is limiting or not. These messages can be used to monitor the operation of this control mode. These messages, INV_Stall_Burst_Model_Temp, INV_Burst_Model_Mode (0 = stall and 1 = high speed), and INV_Limit_Stall_Burst_Model (1 = On/Limiting and 0 = Off) respectively, are in the latest CAN iM-225 DBC file.

2.3 CM200DX Stall Burst Performance

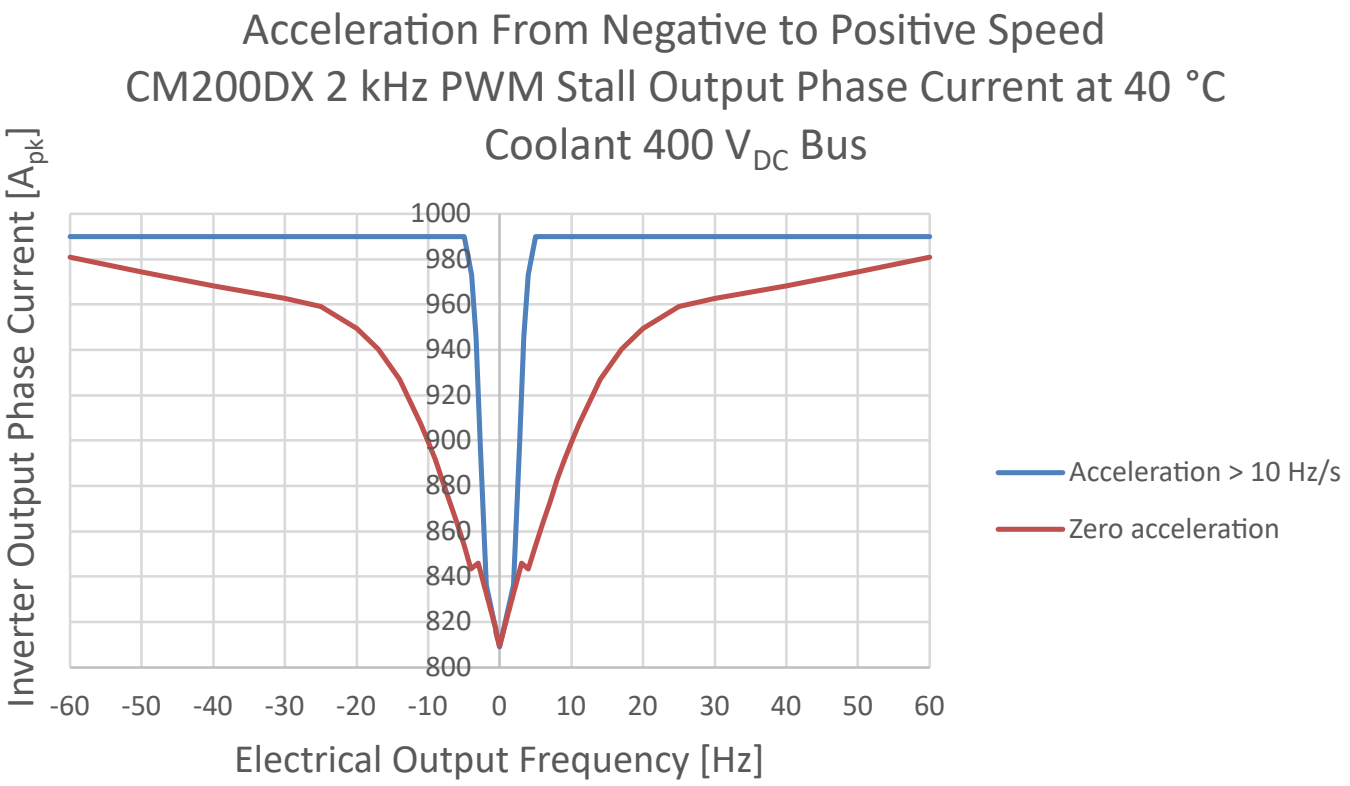


Figure 2: CM200DX Stall Output Phase Current vs Electrical Output Frequency.

The above figure reflects what happens when a vehicle rolls back, then accelerates forward faster than 10 Hz/s. Performance at various constant electrical output frequencies is also shown for comparison. Vehicles rarely have true zero acceleration when full torque is commanded due to driveline slop. Therefore, performance with acceleration better represents typical performance.

Results will vary depending on the initial rotor position because different rotor positions have different distributions of current in each phase. The above figure represents worst case initial rotor position. Sometimes rolling back slightly when on an incline can help low speed performance by changing the rotor initial position.

If held at zero speed, the steady state current is 809.2 A_{pk} for 120 seconds. Current will reduce further to as low as 223 A_{pk} due to hot spot limiting after 120 seconds.

Note, at zero speed phase current is DC, and thus equivalent to A_{pk}. Motor control tables may often list current in A_{RMS}, thus divided the A_{pk} by $\sqrt{2}$ to get the appropriate A_{RMS} for use in lookup tables where necessary.

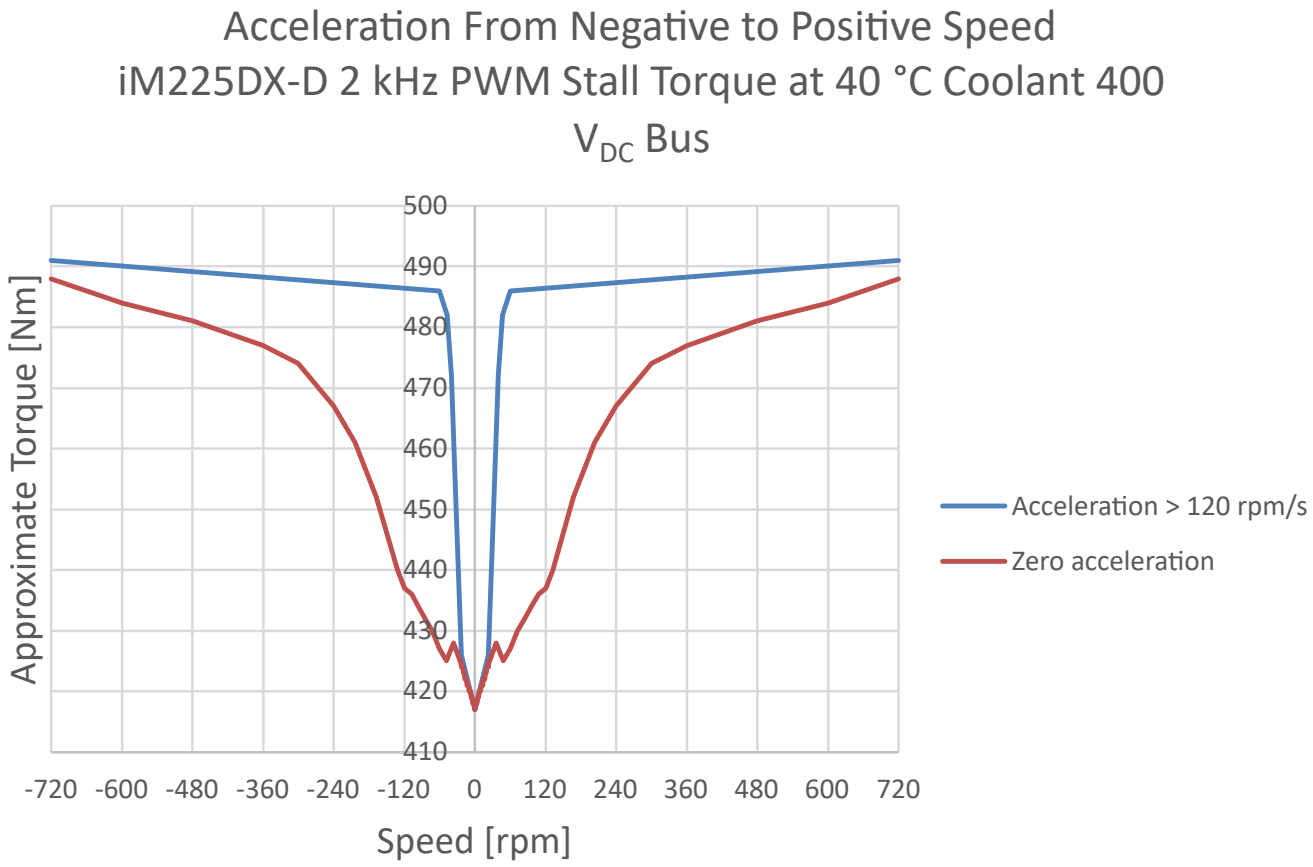


Figure 3: iM225DX-D Stall Torque vs Speed.

2.4 CM200DZ Stall Burst Performance

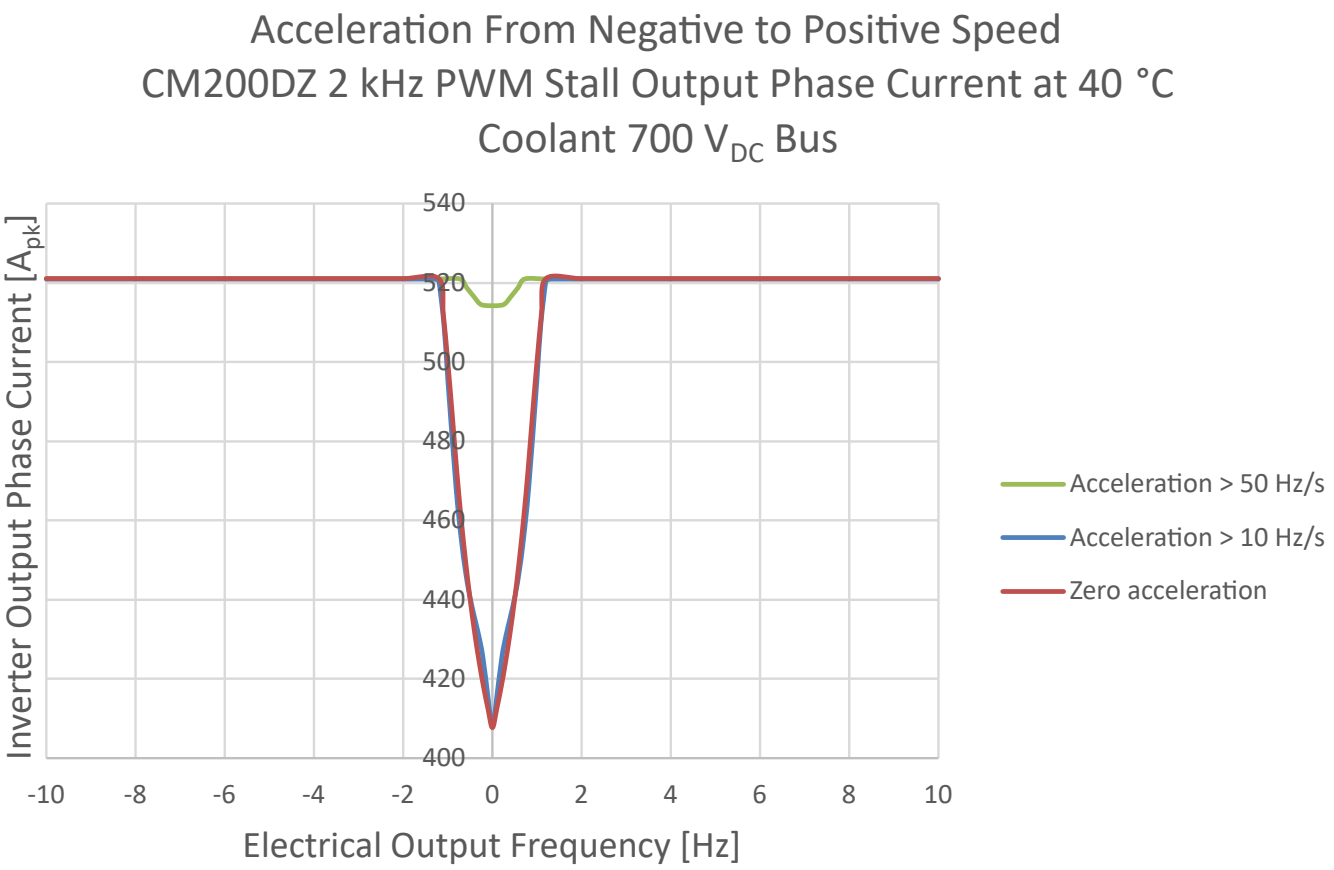


Figure 4: CM200DZ Stall Output Phase Current vs Electrical Output Frequency.

The above figure reflects what happens when a vehicle rolls back, then accelerates forward faster than 10 Hz/s and 50 Hz/s. Performance at various constant electrical output frequencies is also shown for comparison. CM200DZ performance when accelerating can be better than constant output frequency performance. Rolling back and accelerating forward minimizes duration at zero speed resulting in better performance while passing through zero speed. Rolling back and ramping torque during the rollback results in better performance because torque is ramping while spinning which results in a lower initial temperature when entering zero speed. Power module junction temperature rises faster and higher when torque is slowly ramped at zero speed instead of when motor is spinning. Vehicles rarely have true zero acceleration when full torque is commanded due to driveline slop. Therefore, performance with acceleration better represents typical performance.

Results will vary depending on the initial rotor position because different rotor positions have different distributions of current in each phase. The above figure represents worst case initial rotor position.

If held at zero speed, the steady state current is 407.7 A_{pk} for 47 seconds. Current will reduce further to as low as 272 A_{pk} due to hot spot limiting after 47 seconds.

Note, at zero speed phase current is DC, and thus equivalent to A_{pk}. Motor control tables may often list current in A_{RMS}, thus divided the A_{pk} by $\sqrt{2}$ to get the appropriate A_{RMS} for use in lookup tables where necessary.

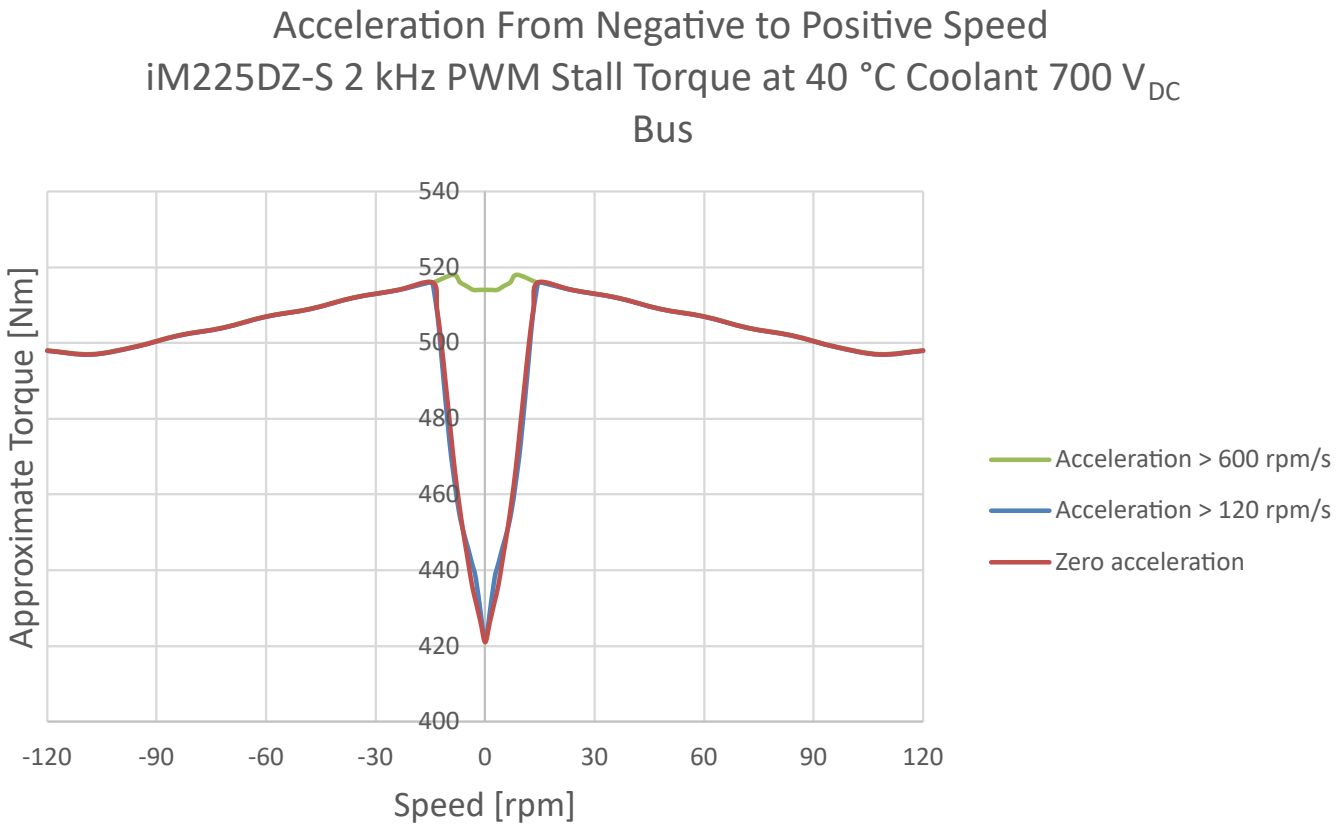


Figure 5: iM225DZ-S Stall Torque vs Speed.

2.5 CM350DZ Stall Burst Performance

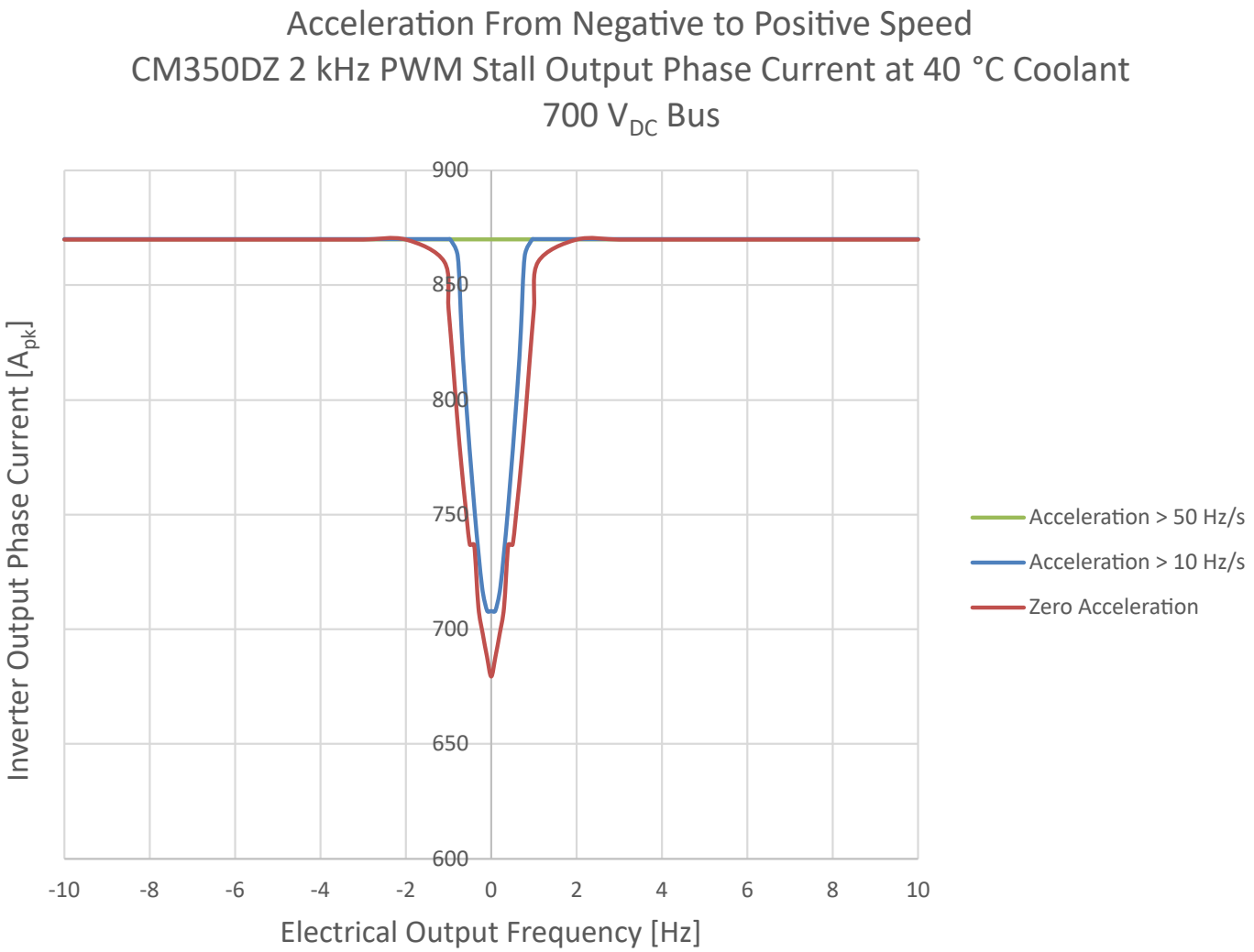


Figure 6: CM350DZ Stall Output Phase Current vs Electrical Output Frequency.

The above figure reflects what happens when a vehicle rolls back, then accelerates forward faster than 10 Hz/s and 50 Hz/s. Performance at various constant electrical output frequencies is also shown for comparison. CM350DZ performance when accelerating can be better than constant output frequency performance. Rolling back and accelerating forward minimizes duration at zero speed resulting in better performance while passing through zero speed. Rolling back and ramping torque during the rollback results in better performance because torque is ramping while spinning which results in a lower initial temperature when entering zero speed. Power module junction temperature rises faster and higher when torque is slowly ramped at zero speed instead of when motor is spinning. Vehicles rarely have true zero acceleration when full torque is commanded due to driveline slop. Therefore, performance with acceleration better represents typical performance.

Results will vary depending on the initial rotor position because different rotor positions have different

distributions of current in each phase. The above figure represents worst case initial rotor position.

If held at zero speed, the steady state current is 679.3 A_{pk} for 106 seconds. Current will reduce further to as low as 655 A_{pk} due to hot spot limiting after 106 seconds.

Note, at zero speed phase current is DC, and thus equivalent to A_{pk}. Motor control tables may often list current in A_{RMS}, thus divided the A_{pk} by $\sqrt{2}$ to get the appropriate A_{RMS} for use in lookup tables where necessary.

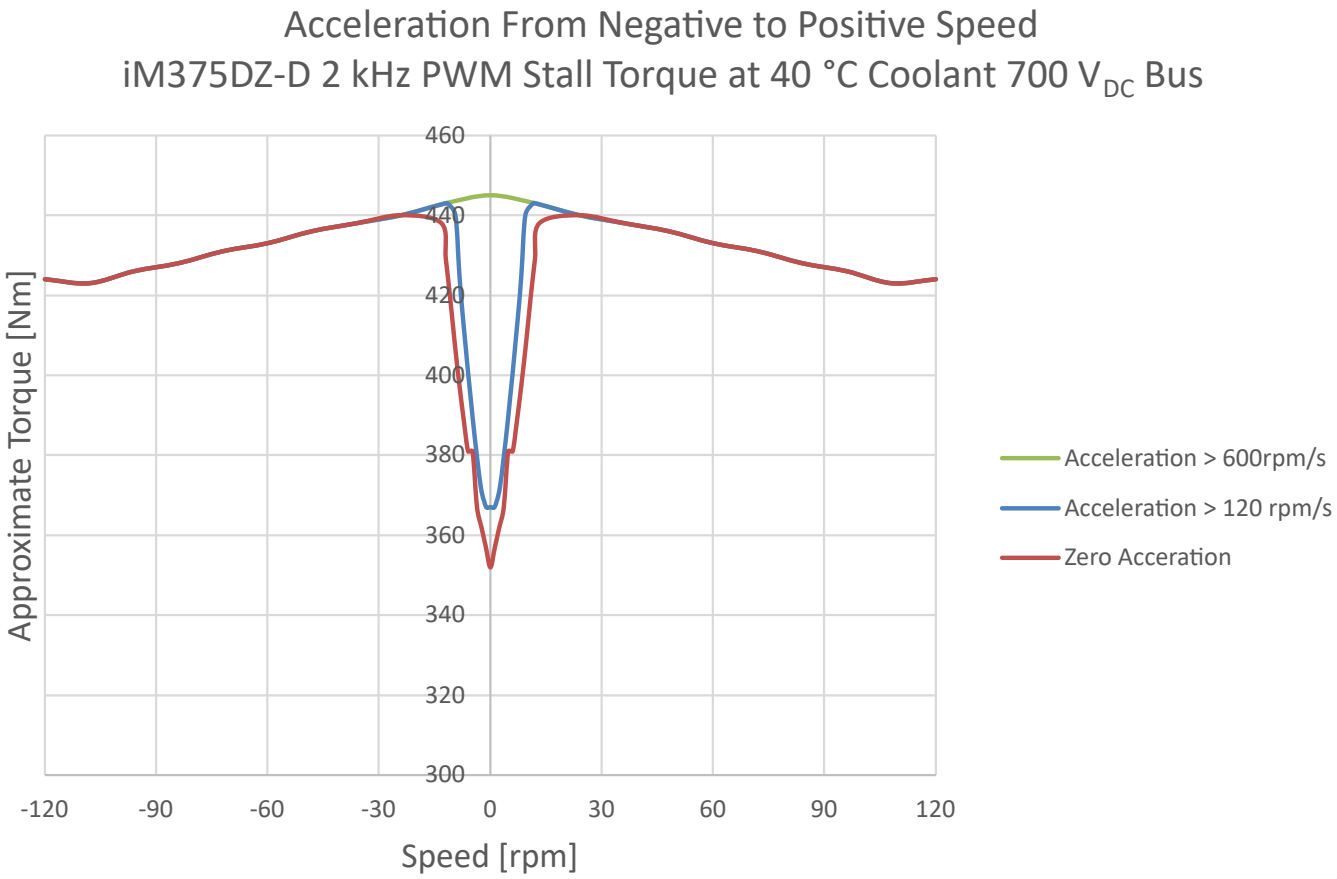


Figure 7: iM375DZ-D Stall Torque vs Speed.

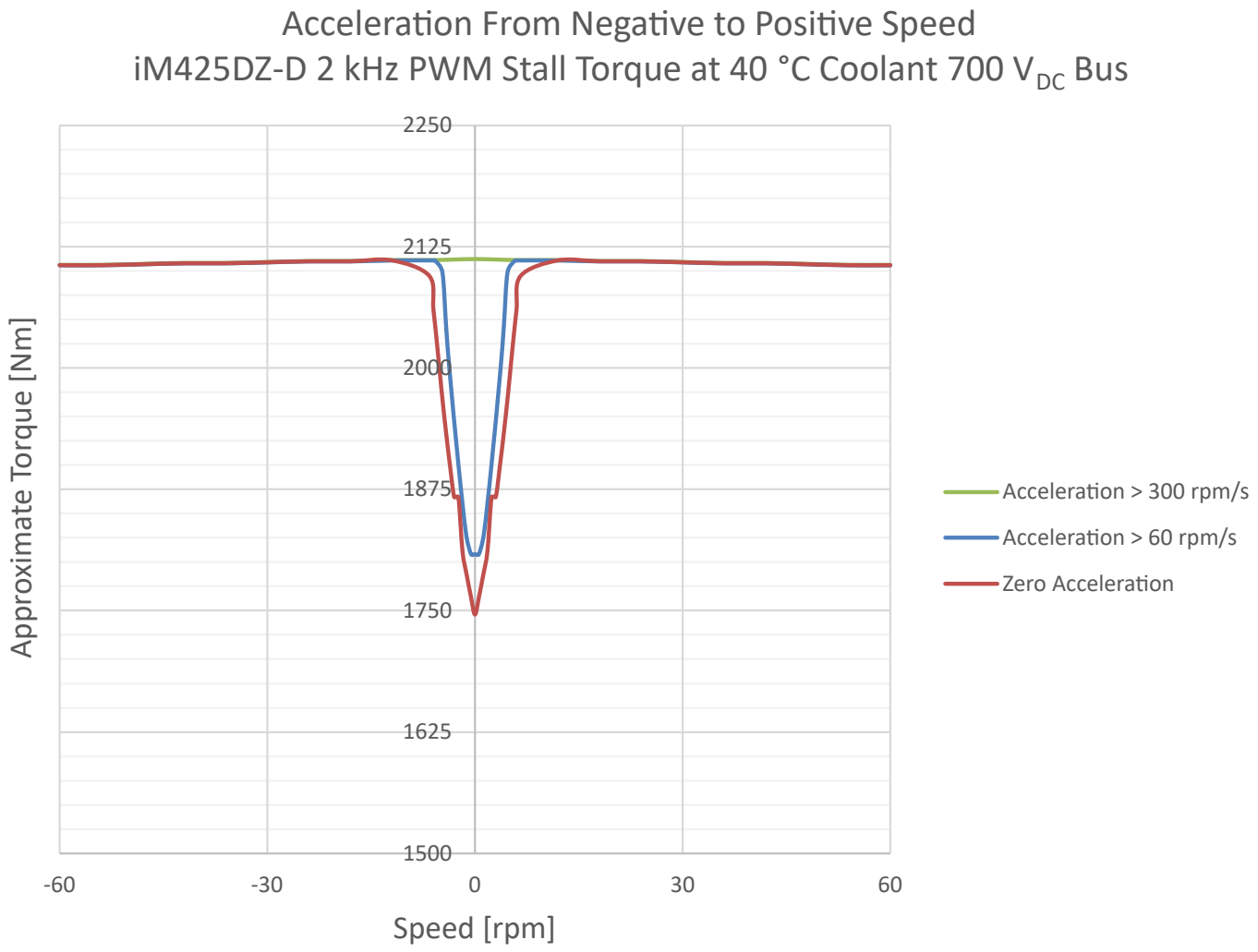


Figure 8: iM425DZ-D Stall Torque vs Speed.

3 Motor Over-temperature Torque Reduction

For Integrated Modules (iM, iDM, & eDM) the motor over-temperature torque reduction and over-temperature limit is 60% of the EEPROM settings while in stall. This is hard coded in 652E firmware and later. The purpose of this is to protect the motor from overheating. Motor temperature ratings are lower while in stall due to poor motor thermal cooling and poor motor temperature measurement while in stall. The CM200DX inverter exits stall at an electrical output frequency of 60Hz or greater. The CM200DZ, CM350DZ, and CM350SiC inverters exit stall at an electrical output frequency of 33Hz or greater.

The figure below shows the relationship between torque capability and motor temperature with Mtr_OverTemp_Limit_EEPROM_(C)_x_10 set to 1500, Zero_Torque_Temp_EEPROM_(C)_x_10 set to 1400, and Full_Torque_Temp_EEPROM_(C)_x_10 set to 1000 as an example.

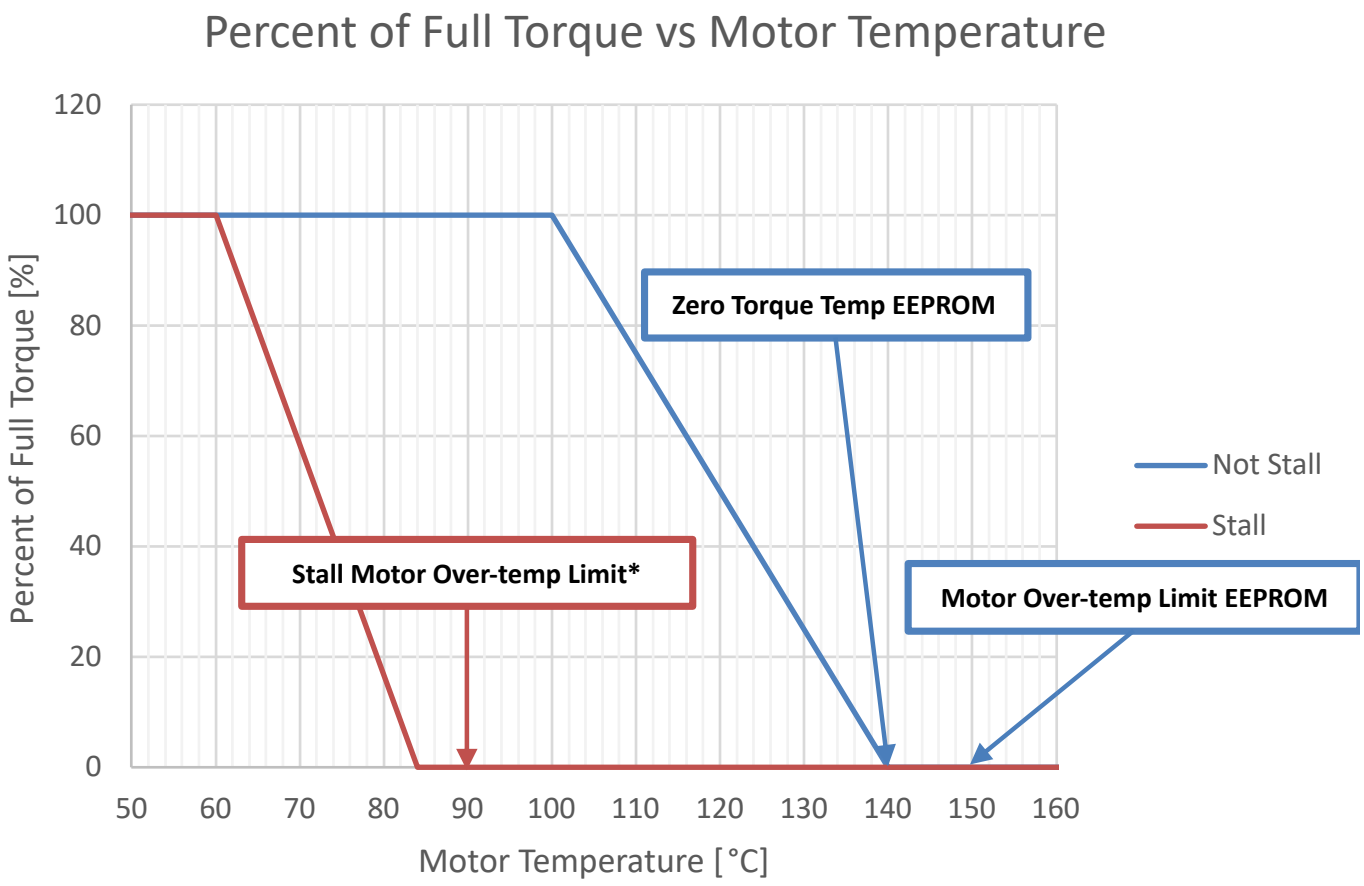


Figure 9: Stall vs Not Stall Torque Reduction Due to Motor Temperature.

* Stall Motor Over-temp Limit is 60% of Motor Over-temp Limit EEPROM only in 652E and 652F firmware. In prior firmware, 6530 firmware and newer firmware Stall Motor Over-temp Limit is the same as Motor Over-temp Limit EEPROM.

It is recommended to use 6530 firmware or newer instead of 652E or 652F. The lower Motor Over-temperature limit in 652E and 652F will show motor over temperature faults even when the motor is not in danger of overheating. This is because a lower temperature limit is only needed when the heat is from stall operation. However, the fault detection for motor overtemperature is only based on current motor temperature and does not know if the heat was generated from stall operation or from operation outside of stall.

4 Revision History

Version	Description of Versions / Changes	Updated By	Date
0.1	Initial version.	Christian Tigges Andrew Louie	7/16/2022
0.2	Added CM200DX stall performance information.	Andrew Louie	1/10/2023
0.3	Added CM200DZ stall performance information. Added CM350DZ stall performance information.	Andrew Louie Christian Tigges	3/27/2023
0.4	Corrected CM200DX stall performance values and plots. Corrected CM200DZ stall performance values and plots. Corrected CM350DZ stall performance values and plots.	Andrew Louie	5/23/2023
0.5	Updated CM200DX stall performance values and plots for 652E firmware and later.	Andrew Louie	3/6/2024
0.6	Add information on changes to Motor Over-temperature Torque Reduction while in stall for 652E firmware and later.	Andrew Louie	6/10/2024